

Sell It Before You Make It

Revolutionizing E-Commerce with Personalized AI-Generated Items

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PerFusion aligns diffusion models with merchants' group-level personalized preferences.

A PARADIGM SHIFT

Generate -> sell -> manufacture on demand

AIGI removes the physical prototype from the critical path and lets merchants test demand before committing inventory.

(a) Traditional E-Commerce Procedure

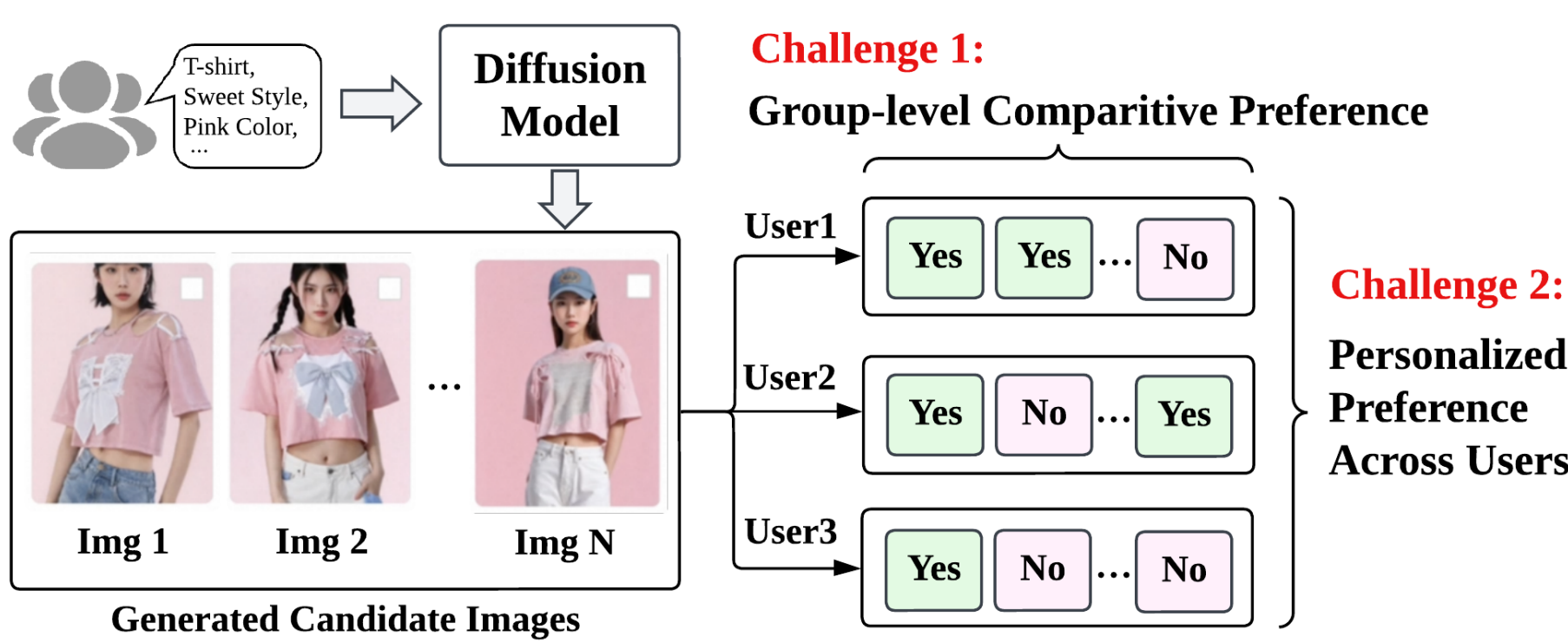


(b) Innovative Procedure with AIGI



Business mode. Merchants generate photorealistic designs first. Production starts only after customer orders reach a threshold - reducing inventory risk and shrinking launch cycles from months to days.

THE SCIENTIFIC GAP



01 | GROUP-LEVEL

Selections are made by comparing an entire candidate set - not isolated positives or pairs.

02 | PERSONALIZED

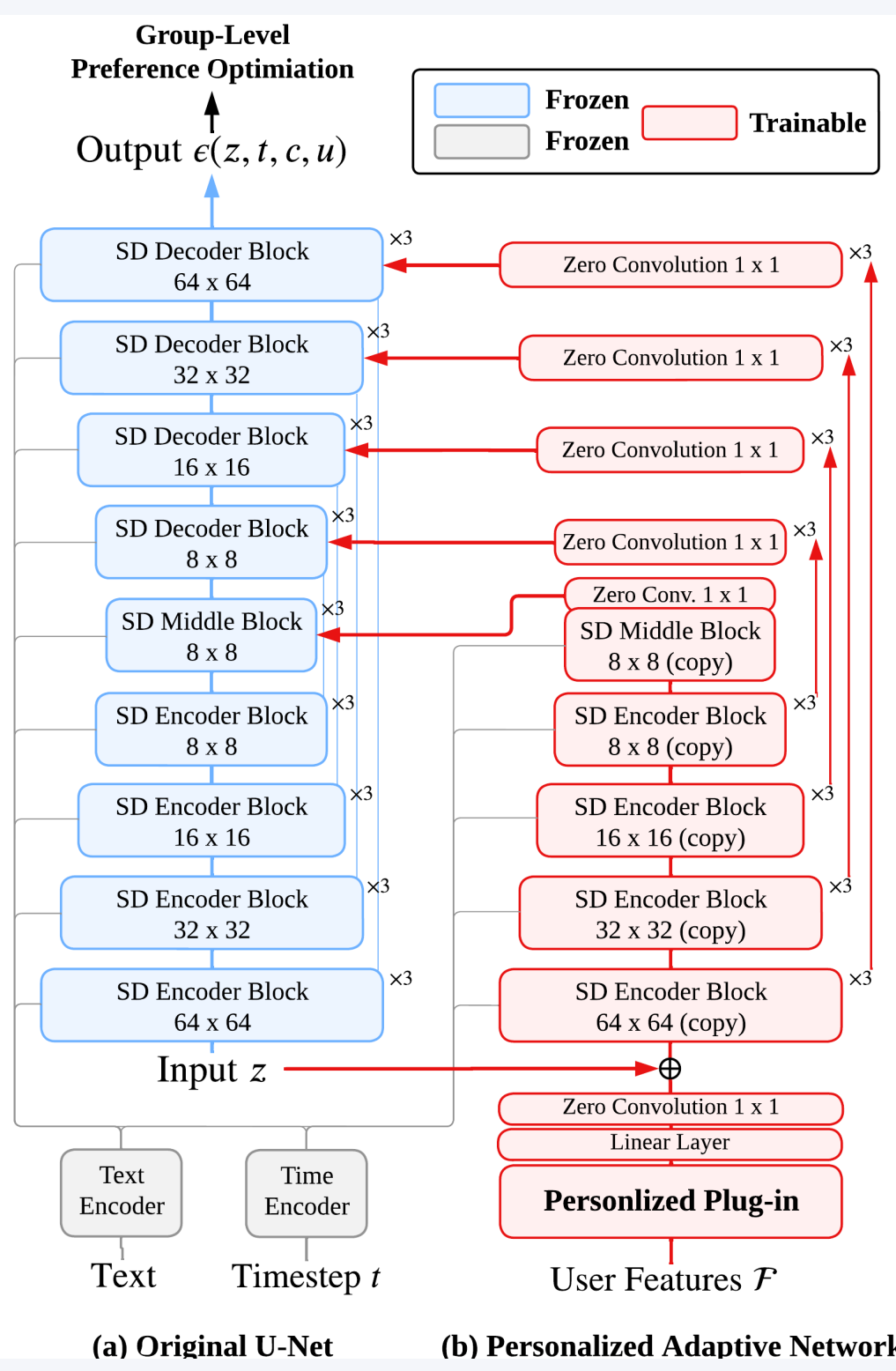
The same prompt and image set can trigger sharply different choices across merchants.

OUR ANSWER: PERFUSION

A personalized reward model plus a personalized adaptive diffusion network, trained with a group-level preference objective. The result is the first industrial-scale deployment of personalized text-to-image generation for item design.

PERFUSION

Two components learn who the merchant is, which candidates they prefer, and how to steer generation accordingly.



PERFUSIONRM

Feature-crossing personalized plug-ins inject merchant representations into both CLIP towers. A group-wise ranking loss learns preference over all candidates.

Output: personalized reward

PERSONALIZED GENERATOR

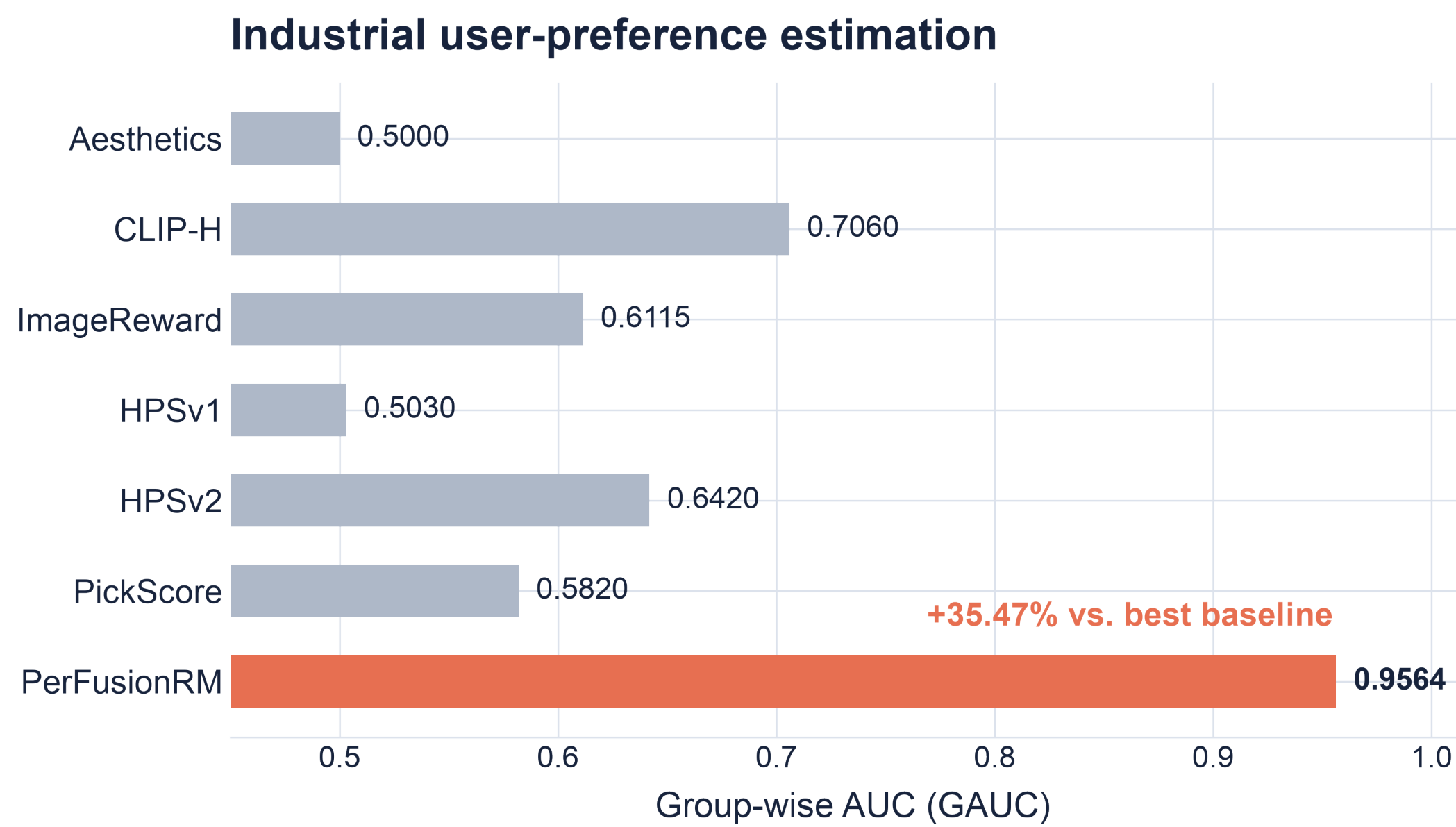
A ControlNet-style adaptive network conditions Stable Diffusion on user features. The original U-Net stays frozen; trainable copies and zero convolutions add personalized signals safely.

Output: preference-aligned images

GROUP-LEVEL PREFERENCE OPTIMIZATION

Replace pairwise Bradley-Terry comparisons with a Plackett-Luce view over positive set P and negative set N. The objective preserves the global candidate context while conditioning on merchant representation u.

OFFLINE EVIDENCE



0.9220

Industrial MAP

0.9564

Industrial GAUC

26.44

Industrial reward

Ablations confirm both pieces matter: removing the personalized adaptive network or the group-level objective degrades personalized generation on PickaPic and Alibaba's industrial dataset.

WORKS IN PRODUCTION

+17.81%

Search CTR

+13.35%

Recommendation CTR

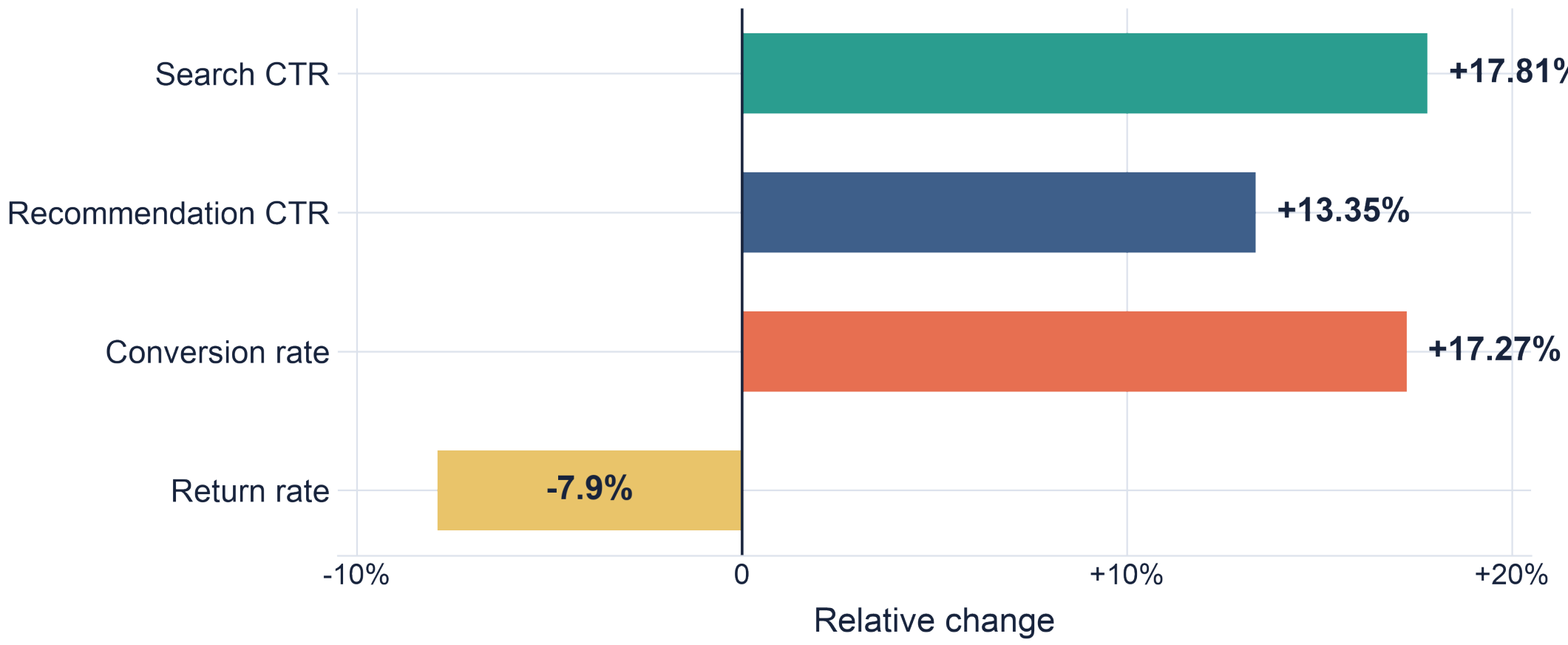
+17.27%

Conversion rate

-7.9%

Return rate

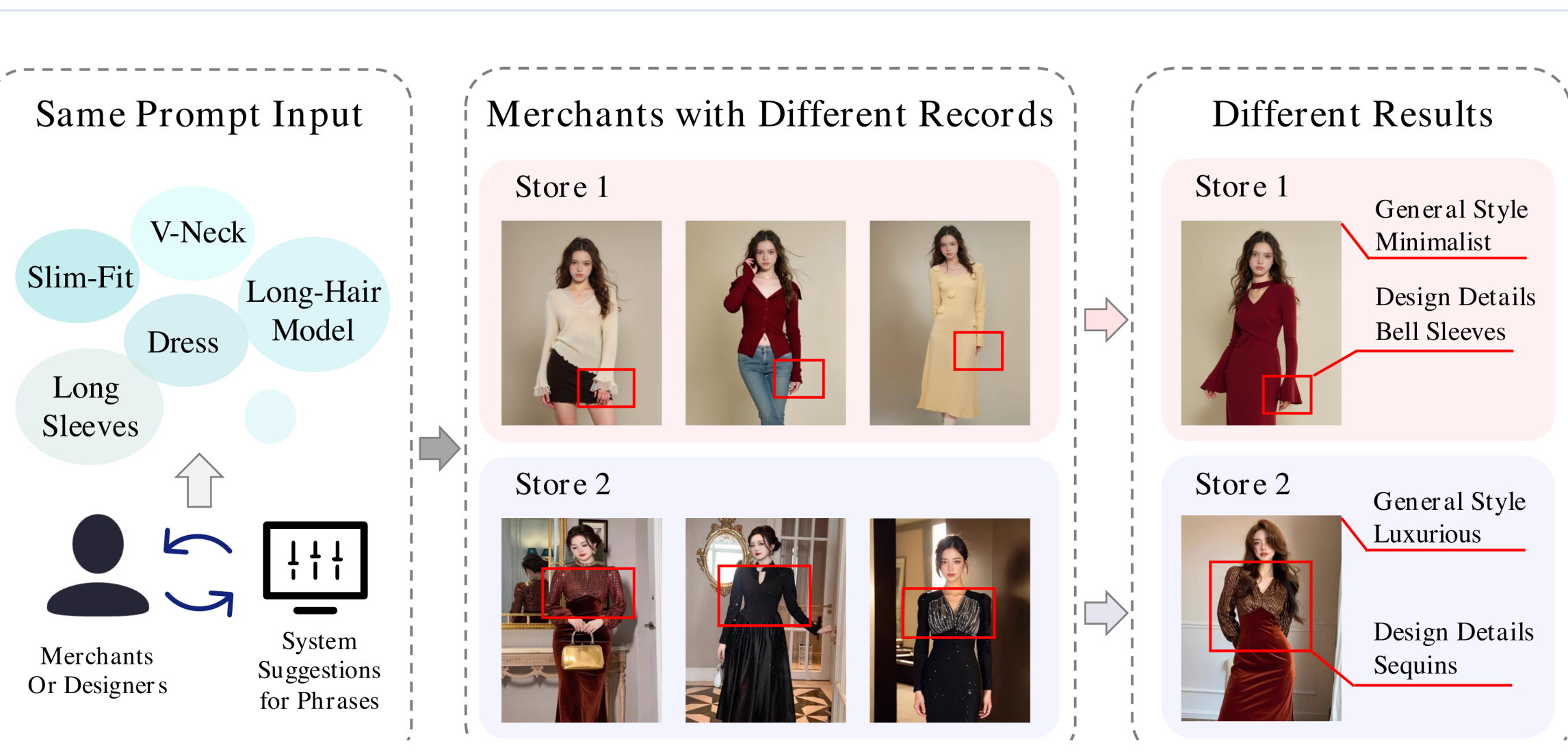
AIGI vs. human-designed items in the same stores



PRODUCT LIFECYCLE

- 1 AI-assisted design**
Generate and select photorealistic items.
- 2 Demand aggregation**
List designs and collect customer orders.
- 3 Manufacture**
Trigger bulk production at a set threshold.

SAME PROMPT, DIFFERENT TASTE



SIX-MONTH DEPLOYMENT

2.5x

Stores using AIGI

2.81x

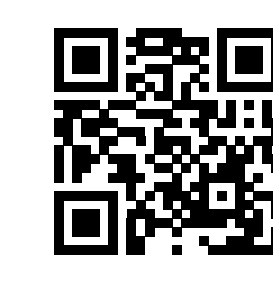
Monthly GMV / store

2.71x

New products / month

TAKEAWAY

PerFusion turns personalized generation into a viable commerce loop: design digitally, validate demand in-market, then manufacture what customers already chose.



SCAN FOR PAPER